COMPUTER ENGINEERING

QUEZON CITY

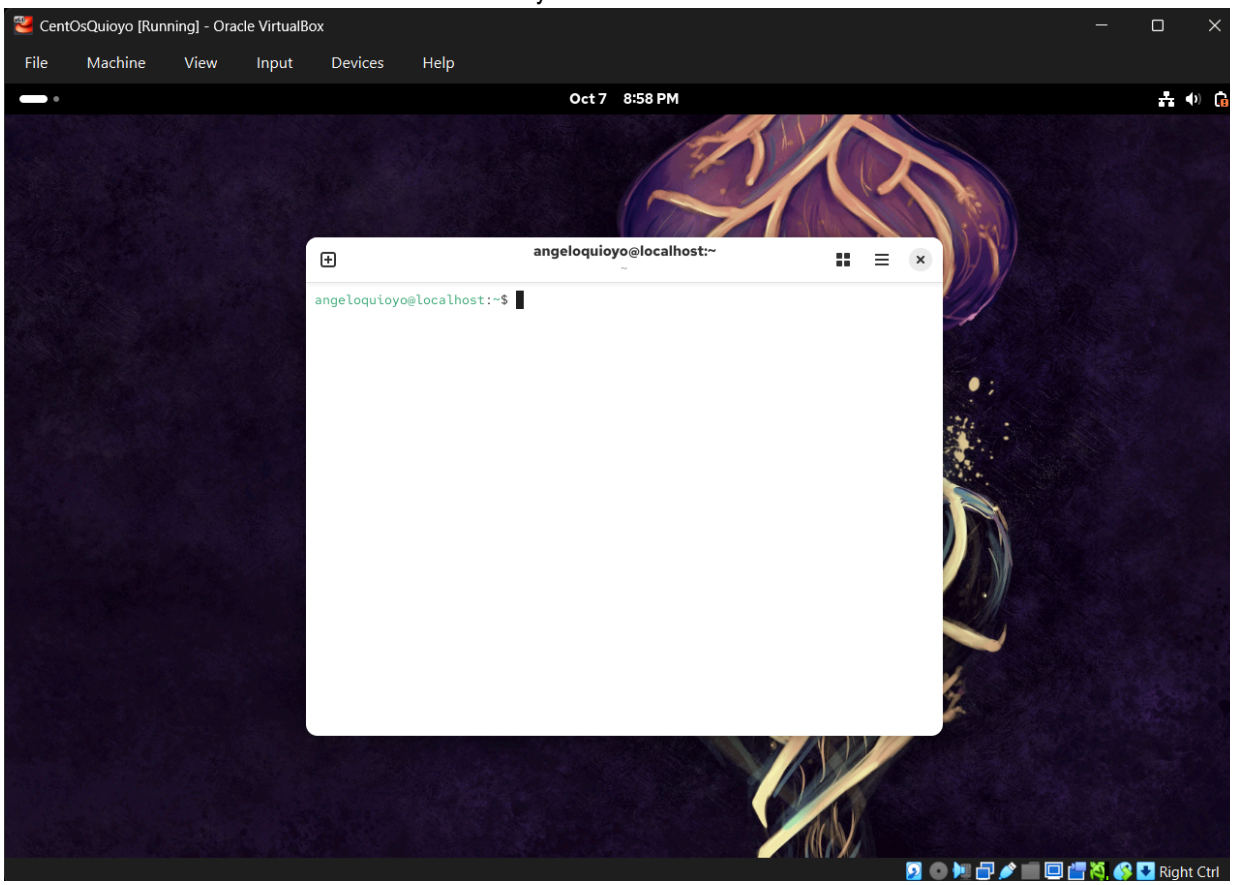
COURSE CODE/TITLE: CPE212 - Automating Server Management		**SECTION:**	CPE31S2
DATE PERFORMED:	10/7/2026	**DATE SUBMITTED:** 10/7/2026	
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ACTIVITY NO. 3 Install SSH server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS 9 Stream here: https://www.centos.org/download/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.

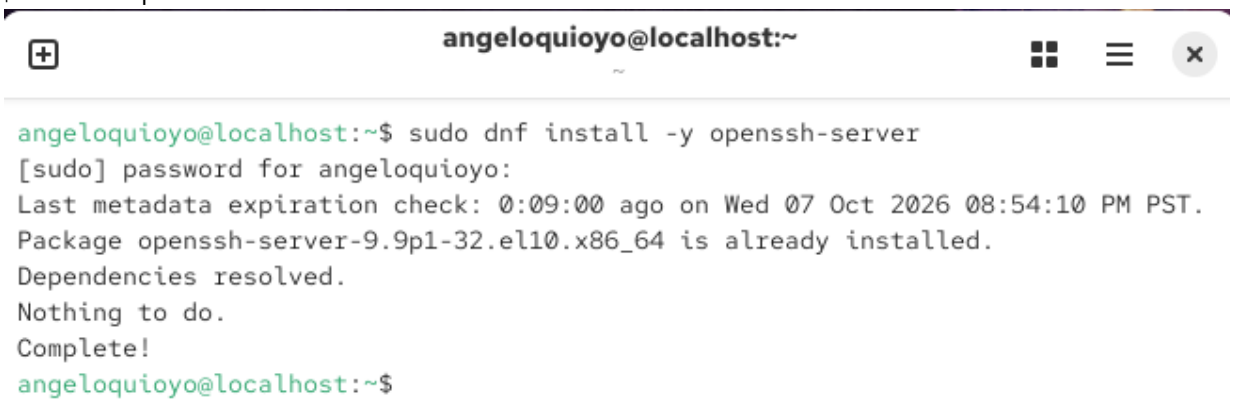




Task 2: Install the SSH server package openssh

1. Install the ssh server package openssh by using the dnf command:

$ dnf install openssh-server



2. Start the sshd daemon and set to start after reboot:

$ systemctl start sshd

$ systemctl enable sshd

```
angeloquioyo@localhost:~$ sudo systemctl start sshd
angeloquioyo@localhost:~$ sudo systemctl enable sshd
angeloquioyo@localhost:~$
```

3. Confirm that the sshd daemon is up and running:
$ systemctl status sshd

```
angeloquioyo@localhost:~$ sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Wed 2026-10-07 21:02:25 PST; 4min 6s ago
 Invocation: 08a25ff353344ec781bd656115c1ba6b
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 1130 (sshd)
      Tasks: 1 (limit: 16971)
     Memory: 1.7M (peak: 1.9M)
        CPU: 14ms
     CGroup: /system.slice/sshd.service
             └─1130 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 07 21:02:25 localhost.localdomain systemd[1]: Starting sshd.service - OpenS>
Oct 07 21:02:25 localhost.localdomain sshd[1130]: Server listening on 0.0.0.0 p>
Oct 07 21:02:25 localhost.localdomain systemd[1]: Started sshd.service - OpenSS>
Oct 07 21:02:25 localhost.localdomain sshd[1130]: Server listening on :: port 2>
lines 1-17/17 (END)
```

4. Open the SSH port 22 to allow incoming traffic:
$ firewall-cmd --zone=public --permanent --add-service=ssh
$ firewall-cmd --reload

```
angeloquioyo@localhost:~$ sudo firewall-cmd --zone=public --permanent --add-serv
ice=ssh
Warning: ALREADY_ENABLED: ssh
success
angeloquioyo@localhost:~$ sudo firewall-cmd --reload
success
```

5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:
$ systemctl reload sshd

```
-------
angeloquioyo@localhost:~$ sudo systemctl reload sshd
angeloquioyo@localhost:~$
```

Task 3: Copy the Public Key to CentOS

1. Make sure that ssh is installed on the local machine.

```
Generating public/private rsa key pair.
Enter file in which to save the key (/home/angeloquioyo/.ssh/id_rsa):
Created directory '/home/angeloquioyo/.ssh'.
Enter passphrase for "/home/angeloquioyo/.ssh/id_rsa" (empty for no passphrase):

Enter same passphrase again:
Your identification has been saved in /home/angeloquioyo/.ssh/id_rsa
Your public key has been saved in /home/angeloquioyo/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:BNcD3DBkpOsDZf+FNOZxNlQbQGJUn0TdZsfJgxt6gnQ angeloquioyo@localhost.locald
omain
The key's randomart image is:
+---[RSA 4096]----+
|      .+O*+o=+Boo|
|       =.++E = BB|
|      + o B.= *oo|
|     o + = O +   |
|    . . S o +    |
|     o   . .     |
|      o   .      |
|       .         |
|                 |
+----[SHA256]-----+
angeloquioyo@localhost:~$
```

2. Using the command ssh-copy-id, connect your local machine to CentOS.

```
angeloquioyo@localhost:~$ ssh-copy-id angeloquioyo@10.0.2.15
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.
ED25519 key fingerprint is SHA256:o3Zu+AkpW3p9mCgZ6KbNeOVwMebs9Rn5ISBlP7hNToI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
angeloquioyo@10.0.2.15's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'angeloquioyo@10.0.2.15'"
and check to make sure that only the key(s) you wanted were added.

angeloquioyo@localhost:~$
```

3. On CentOS, verify that you have the authorized_keys.

```
angeloquioyo@localhost:~$ cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQC2a8IdEySNUqLXPswOeDm0ivWu9vn3ztGVTO6a6qlb
IFJJLnFPAohnyekCepij2twWW+FplGJFeKuiuoi4zIEqUalfQm0+5iUjmXfrWB97vgYKYrjJBK1oIYl2
+UC6ZpiLcGUutr6nWPwZYPIt31EeeFNJZCirmW3praWtemYZ1ov5OT7y7kw6sgWms0ENFGIC87jSAfmS
h++UYWShgtvTJ2UNZJL5ctfTSbLCM7KfVQjPsDQWZrWEGQNOAKDmMpOqoGX9tG0Fhp/CGE00jStNcedk
v3ruRxovyOpNQCedBNGWbiNb6yVkLmXcayEWAFxoFJZICqi/oUR81u7Fgu5Bcm5BeePYB4T/6cJVU2mg
HMWPaofXeX/gR+XzUjxarjx/eP1+Hd6G/qdxrHBGpWlMmQsNpkbC4ARM6ASNrUO2FjGpejH2VKJ1IK4a
8X44PCBpa/sssKU7Pe/RiwBQo6GPJKI7AtITb1ZVjLpLRqP1V1UXXhZmZBD2vA81taXIt5ub1y3j4tEf
QIeD0zj7N1MhHjq/FLib/s/uV8WJugNfTeDjEI9oc8rOs2cj4gWJZfvCgURvAs/heCl/FOqOPVgjUdTx
5mD8QVDku+wMRmZbyGAj27M0ihZv8UfjApveUMW2TyyK98avHYqVXz1lv/X0t4LESNC2TTG8hHuFOGzz
qQ== angeloquioyo@localhost.localdomain
angeloquioyo@localhost:~$
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.

```
angeloquioyo@localhost:~$ ssh angeloquioyo@10.0.2.15
Web console: https://localhost:9090/ or https://10.0.2.15:9090/

Last login: Wed Oct  7 21:02:38 2026
angeloquioyo@localhost:~$
```

2. Show evidence that you are connected.

```
angeloquioyo@localhost:~$ whoami
angeloquioyo
angeloquioyo@localhost:~$ hostname
localhost.localdomain
angeloquioyo@localhost:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:4e:e1:39 brd ff:ff:ff:ff:ff:ff
    altname enx0800274ee139
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 85557sec preferred_lft 85557sec
    inet6 fd17:625c:f037:2:a00:27ff:fe4e:e139/64 scope global dynamic noprefixro
ute
       valid_lft 86128sec preferred_lft 14128sec
    inet6 fe80::a00:27ff:fe4e:e139/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
angeloquioyo@localhost:~$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux Distributions?
   - Assess the need for commercial vendor support and enterprise software compatibility. Red Hat offers official commercial backing while Debian operates as a community-driven project. Verify that the distribution supports your targeted hardware architecture like x86_64, AArch64, or MIPS64el.
2. What are the main differences between Debian and Red Hat Linux distributions?
   - Red Hat uses RPM packages with DNF while Debian uses DEB packages with APT. Red Hat operates as a commercial downstream distribution while Debian serves as a community upstream distribution. Debian supports architectures like s390x and MIPS64el while CentOS 8 does not.

Conclusions/Learnings:

- In conclusion, I installed CentOS Stream on VirtualBox with 2 GB RAM and 20 GB storage. Configured OpenSSH, enabled the service using systemctl, and opened port 22 in firewalld. Established passwordless SSH authentication by sending your public key using ssh-copy-id

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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